



FACILITY INFECTION-CONTROL BRIEF

Built-in protection, not **spray-and-repeat.**

How FUZE durable antimicrobial treatment compares to electrostatic spray disinfection — and the everyday chemistry behind the spray — for the materials your patients, residents, and staff touch all day.

Dental practices

Healthcare facilities

Government & multi-unit housing

A bonded treatment vs. a recurring spray event

Electrostatic spraying is a *delivery method* — a charged mist that coats surfaces with a liquid disinfectant. It works while it is wet and until the space is touched again. FUZE is a *durable treatment* that bonds antimicrobial protection into the material itself, so the protection stays put between cleanings instead of evaporating with the mist.

FUZE — durable, built-in

A non-leaching metamaterial permanently bonds into treated textiles and substrates during manufacturing or finishing. It works on contact — no chemical is released into the air or wash water.

- ✓ Applied **once**; protection is engineered into the material
- ✓ **Non-leaching** — nothing sprays off, washes off, or off-gases
- ✓ Stays active **between cleanings**, 24/7
- ✓ F1 tier retains efficacy through **100 wash cycles**
- ✓ No quats, no PFAS, no formaldehyde, no heavy-metal residue
- ✓ No recurring labor, mixing, or per-room respray

Electrostatic spray — topical, temporary

A handheld or backpack unit charges droplets of an EPA-listed liquid disinfectant — most often a quat, hypochlorous acid, or hydrogen peroxide — so they wrap onto surfaces. It is a point-in-time event.

- Must be **reapplied** on a schedule and after each use of the space
- Surface must stay visibly **wet ~10 minutes** to hit label kill claims
- Protection ends once the surface dries or is re-touched
- Leaves a chemical **residue film** on every coated surface
- Recurring chemical cost + staff labor, indefinitely
- Common chemistries carry handler and resistance concerns (next page)

The core difference: electrostatic spraying lowers microbes at the moment it is applied; FUZE lowers the microbial burden the material carries **all day, between cleanings**. They are not the same job — and FUZE does not need a tank, a mist, or a dwell timer.

The comparison — and the chemistry

"Electrostatic" describes the sprayer, not the disinfectant. The active chemistry is what touches your staff, patients, and residents every day.

FACTOR	FUZE TREATMENT	ELECTROSTATIC SPRAY
Duration	Durable; bonded for the life of the wash cycle (up to 100 washes, F1)	Lasts through the wet dwell time only; re-spray to maintain
Mechanism	Non-leaching, contact-based; nothing released into air or water	Topical liquid film that must stay wet to work
Active chemistry	FUZE metamaterial — no quats, PFAS, or formaldehyde	Quats, hypochlorous acid, or hydrogen peroxide
Recurring labor	None after treatment	Ongoing — mixing, spraying, dwell, every cycle
Residue	None — bonded into the fiber/substrate	Chemical film left on coated surfaces
Verification	Third-party tested: ASTM E2149, AATCC 100, ISO 20743	Relies on correct dwell time per spray, every spray
Certifications	EPA registered · OEKO-TEX Class I · bluesign® · PFAS-free	Varies by product; EPA-listed disinfectant required

⚠ Quats — the most common spray chemistry

- ▶ Linked to **occupational asthma** in healthcare staff, with the highest risk during manual mixing and repeat peak exposure.^{4,5}
- ▶ Associated with **contact dermatitis** and skin irritation in frequent handlers.⁵
- ▶ Reviews flag **antimicrobial-resistance** and antibiotic cross-resistance from sub-lethal residue build-up.^{6,7}
- ▶ Persists in the environment and in wastewater.^{6,7}

✓ What FUZE keeps out of the room

- ✓ No quaternary ammonium compounds
- ✓ No PFAS "forever chemicals"
- ✓ No formaldehyde or heavy-metal leaching
- ✓ Nothing aerosolized for staff to inhale
- ✓ OEKO-TEX Standard 100 Class I — safe for direct skin contact

Why it matters for your facility: spraying puts a fresh layer of disinfectant chemistry into the air and onto surfaces every cycle, where staff handle and breathe it. FUZE adds **no recurring chemical load** — the protection is already in the material.

A durable layer under your cleaning protocol

FUZE is not a replacement for required cleaning or terminal disinfection — it works underneath them, holding down the microbial burden on the materials in between. Three ways facilities deploy it:

1

Treat the soft goods

Scrubs, lab coats, dental bibs, privacy curtains, upholstery, bedding, towels and linens — the high-touch textiles a sprayer reaches inconsistently.

2

Cut the respray load

Durable built-in protection reduces how hard each cleaning cycle has to work to manage odor- and deterioration-causing microbes on treated items.

3

Lower the chemical footprint

Less recurring quat spray means less aerosolized chemistry for staff, patients and residents — and a cleaner environmental profile.

Typical treated items in your setting

Dental bibs & towels

Scrubs & lab coats

Privacy curtains

Chair & exam upholstery

Bedding & linens

Resident soft furnishings

Common-area textiles

See it on your own materials

We'll treat a sample of your facility's fabric and return third-party test data (ASTM E2149 / AATCC 100 / ISO 20743) so you can evaluate durable protection against your current spray program — on your goods, not a brochure.

FUZE Biotech · andrew@fuze47.com · fuzefaq.com

Compliance note. FUZE-treated articles are protected under the EPA treated-article exemption. FUZE provides durable antimicrobial protection of the treated material itself (inhibiting odor- and deterioration-causing bacteria, mold and mildew) and is not represented as a stand-alone public-health disinfectant or a substitute for your facility's required surface-disinfection and infection-control protocols. Specific pathogen-reduction data is available under separate technical documentation. FUZE / metamaterial are FUZE Biotech designations for our proprietary non-leaching antimicrobial.

Sources. (1) EMist — "A Guide on What Chemical to Use in Electrostatic Sprayers." (2) CleanLink — "The Case for Electrostatic Sprayers and Hypochlorous Acid." (3) CDC / EPA disinfectant dwell-time guidance (Regency Supply summary). (4) Gonzalez et al., "Asthma among workers in healthcare settings: role of disinfection with quaternary ammonium compounds," *Clinical & Experimental Allergy*, 2014. (5) "Occupational Asthma Caused by Quaternary Ammonium Compounds: A Multicenter Cohort Study," *J. Allergy Clin. Immunol. In Practice*, 2021. (6) "Challenges of quaternary ammonium antimicrobial agents: mechanisms, resistance, persistence and impacts on the microecology," *Science of the Total Environment*, 2024. (7) "Quaternary ammonium compounds of emerging concern: classification, occurrence, fate, toxicity and antimicrobial resistance," *PMC*, 2022.